A Descriptive Revisit of the U.S. Beveridge Curve, 2001–2024

Real-data research pilot

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Abstract

Using a frozen snapshot of 288 paired monthly U.S. labor-market observations from the Bureau of Labor Statistics (BLS), this note compares job-openings rates in months with similar unemployment rates before and after the pandemic transition. In the 3.5–4.5 percent unemployment interval, the mean openings rate was 4.10 percent in 2001–2019 and 5.70 percent in 2022–2024, a descriptive gap of 1.60 percentage points. The 2024-only gap was 0.55 points. This is a reproducibility and workflow pilot of an already known pattern, not a novel or causal finding. Survey measurement, revisions, compositional change, serial dependence, and limited support restrict interpretation.

1 Introduction

The Beveridge curve plots vacancies against unemployment. BLS has long published precisely this comparison and documented outward movements in earlier periods [U.S. Bureau of Labor Statistics, 2013, a]. This note asks a narrow descriptive question: among months with comparable measured unemployment, how different were published U.S. job-openings rates in 2022–2024 relative to 2001–2019? The exercise tests an end-to-end research workflow using real government data, rather than proposing a new economic mechanism or claiming a contribution to the literature.

2 Related Evidence

BLS's earlier Beveridge-curve chart already reports time variation in the vacancy–unemployment relationship [U.S. Bureau of Labor Statistics, 2013]. The live BLS chart uses the same two aggregate rate series [U.S. Bureau of Labor Statistics, a]. Barlevy et al. [2023] analyze changing reasons for Beveridge-curve shifts through 2023, while Kindberg-Hanlon and Girard [2024] study the pandemic shift with state data and a search model. Their scope and interpretation are much deeper than our two-period summary. Recent BLS work examines whether unusually high JOLTS openings after 2020 reflect reporting or processing error [Akinyooye et al., 2026]. Our arithmetic comparison neither supersedes those analyses nor resolves the mechanisms behind the joint movement. Its contribution is limited to an auditable real-data workflow demonstration.

3 Data and Measurement

We retrieved the seasonally adjusted civilian unemployment rate (CPS series LNS14000000) and total nonfarm job-openings rate (JOLTS series JTS000000000000000JOR) directly from BLS Public Data API version 2.0 [U.S. Bureau of Labor Statistics, 2014, a]. The snapshot contains monthly observations from January 2001 through December 2024: 288 months paired by year and month. The raw JSON bodies, timestamps, request windows, hashes, and acquisition script are in the research package. BLS describes a job opening as an unfilled position with available work, a start possible within 30 days, and active external recruitment [U.S. Bureau of Labor Statistics, c]. The openings rate divides openings by employment plus openings. Both values are published aggregate estimates, not simulated observations or individual-level records.

The baseline is January 2001–December 2019 (228 months); the comparison is January 2022–December 2024 (36 months). We exclude the 24 months of 2020–2021 from the comparison to avoid mixing acute pandemic and reopening observations into either window. This is an ex-post descriptive choice. No observations are missing in the retrieved paired series. BLS estimates can be revised; the frozen snapshot defines the replication target. BLS published material is public domain with requested attribution [U.S. Bureau of Labor Statistics, b].

4 Descriptive Design

Let U_t and V_t denote published unemployment and openings rates. For an unemployment interval I, we compute

$$\Delta(I) = \text{mean}(V_t \mid t \in 2022{:}2024, U_t \in I) - \text{mean}(V_t \mid t \in 2001{:}2019, U_t \in I).$$

The primary interval is $I = [3.5, 4.5]$ percentage points, chosen to create common support across the two windows. A secondary interval $[4.0, 5.0]$ checks sensitivity to the support definition. We also compare 2024 alone with the baseline in the primary interval. These are finite-sample arithmetic summaries. They do not estimate a treatment effect or identify a structural Beveridge-curve shift. We do not attach conventional independent-month standard errors because the monthly points are serially related and selection into an unemployment interval is endogenous to the macroeconomy.

5 Results

Figure 1 plots every included month; the transition years are omitted from the display. The table reports means, with numbers of months in parentheses. In the primary interval, the baseline contains 48 months and the comparison contains 35. Their average openings rates differ by 1.60 percentage points. The broader shifted interval gives a 1.21-point difference but contains only 8 comparison months; it is a weak sensitivity check.

Table 1: Mean job-openings rate in comparable unemployment intervals (percent)

Unemployment interval	2001–19	2022–24	Difference
3.5–4.5%	4.10 (48)	5.70 (35)	1.60
4.0–5.0%	3.57 (75)	4.79 (8)	1.21

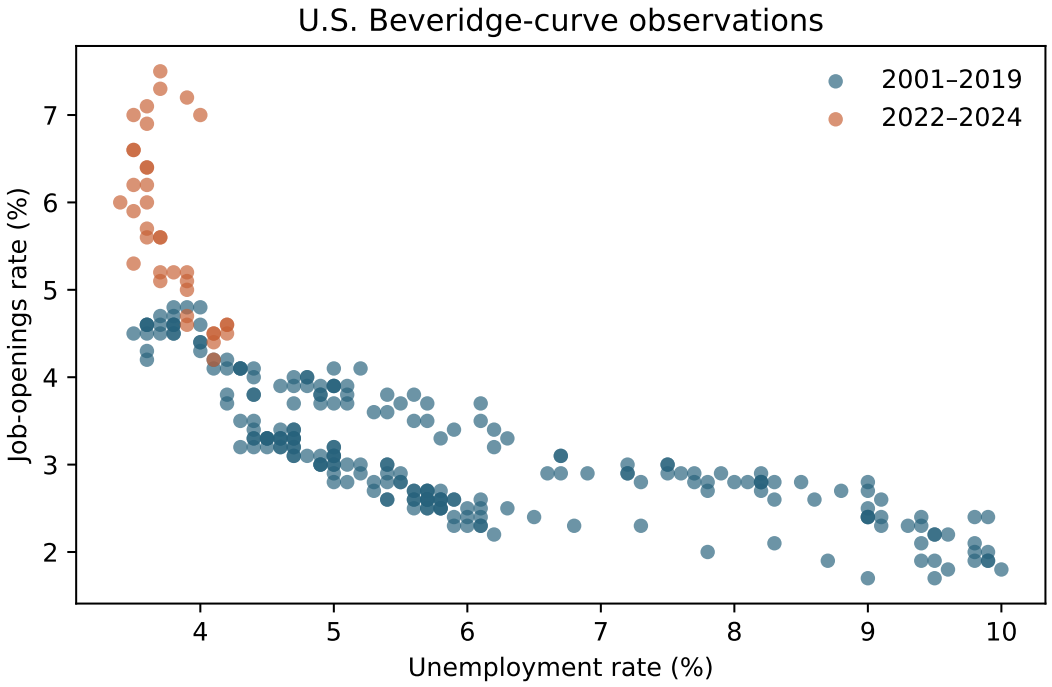


Figure 1: Published seasonally adjusted monthly U.S. rates. Each dot is one month. The graph is descriptive and does not fit a structural curve.

6 Robustness and Failed Persistence Hypothesis

The same primary-interval calculation for 2024 alone uses 12 months. Its mean openings rate was 4.65 percent, only 0.55 points above the 2001–2019 conditional mean. Thus the pooled post-2021 difference was much larger than the difference in the final observed year. A trial claim that the 2022 gap remained equally large through 2024 was discarded in the experiment ledger.

7 Limitations and Interpretation

The observed conditional difference is not evidence that matching efficiency worsened, a policy caused a shift, or a stable structural curve moved. Vacancy and unemployment measures cover different survey populations; industry mix, recruiting intensity, wages, labor supply, and business-cycle composition can all change their joint distribution. The IMF analysis expresses vacancies relative to the labor force, whereas our BLS openings rate divides openings by employment plus openings; their magnitudes are therefore not directly comparable [Kindberg-Hanlon and Girard, 2024]. BLS's JOLTS estimates have sampling and nonsampling error, and histories are revised [U.S. Bureau of Labor Statistics, 2024]. Conditioning on a narrow unemployment interval does not make the unemployment distributions within that interval identical, so the mean gap is not a fixed-unemployment vertical distance. It also does not remove the other confounders. The $[4.0, 5.0]$ comparison has just eight post-period months. Annual averages also conceal within-year dynamics. We do not infer statistical significance or extrapolate the 2022–2024 pattern past December 2024. The old and new windows are deliberately different calendar eras, so any difference is descriptive of these selected months only.

8 Conclusion

This pilot reproduces a familiar feature of published U.S. labor-market aggregates: at similar recorded unemployment rates, openings were higher on average in 2022–2024 than in 2001–2019. The gap had narrowed substantially by 2024. The deliverable is a reproducible real-data descriptive note and a tested research package, not a new causal or top-journal contribution.

9 Reproducibility Appendix

The public BLS API was queried in three non-overlapping windows (2001–2010, 2011–2020, 2021–2024). Frozen response bodies, SHA-256 hashes, and acquisition logs are in the `data/` directory. Running the analysis script from the project root regenerates the panel, results file, table, and figure. The replication configuration specifies a clean isolated replay. The acquisition script refuses to overwrite a prior raw snapshot. This appendix documents computational reproduction of the recorded vintage, not future BLS releases.

References

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